Transformation of Multispectral Data to Quasi-Hyperspectral Data Using Convolutional Neural Network Regression

Subir Paul and D. Nagesh Kumar

***Abstract*— Hyperspectral (HS) data are proven to be more resourceful compared to multispectral (MS) data for object detection, classification, and several other applications. However, absence of any space-borne HS sensor since 2017, which can provide open-source data with global coverage, and high cost and limited obtainability of airborne sensors-based images limit the use of HS data. Transformation of readily available MS data into quasi-HS data can be a feasible solution for this issue. In this article, we propose the use of convolutional neural network regression (CNNR), a deep learning-based algorithm, for MS (i.e., Landsat 7/8) to quasi-HS (i.e., quasi-Hyperion) data transformation. The proposed CNNR model is compared with the existing pseudo-HS image transformation algorithm (PHITA), a simple linear model [i.e., stepwise linear regression (SLR)], and a nonlinear modeling approach [i.e., support vector regression (SVR)] by evaluating the quality of the quasi-Hyperion data. Contrary to these existing and simple models, the proposed CNNR model has the added advantage of utilizing deep learning-based spectral–spatial features for MS to quasi-HS data transformation through regression-based nonlinear modeling. Different statistical metrics are calculated to compare each band's reflectance values as well as spectral reflectance curve of each pixel of the quasi-Hyperion data with that of the original Hyperion data. The developed models and generated quasi-Hyperion data are also evaluated with application to crop classification. Analyzing the results of all the experiments, it is evident that CNNR model is more efficient compared to PHITA, SLR, and SVR in creating the quasi-Hyperion data and this transformed data are proven to be resourceful for crop classification application. The proposed CNNR model-based MS to quasi-HS data transformation approach can be used as a viable alternative for different applications in the absence of original HS images.**

***Index Terms*— Convolutional neural network regression (CNNR), crop classification, hyperspectral (HS) data, multispectral (MS) data, quasi-HS data.**

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I. Introduction

REMOTE sensing data are very important for continuous spatial and temporal monitoring of Earth and its different processes. Remote sensing images are generally characterized based on spectral, spatial, temporal, and radiometric resolutions. Hyperspectral (HS) data provide finer spectral information with hundreds of narrow bands (i.e., ∼1–10-nm spectral resolution) as compared with coarser spectral bands (i.e., ∼30–100-nm spectral resolution) of multispectral (MS) data. Large number of narrow spectral bands provides abundant spectral information and helps in identifying different Earth surface features and land cover types (e.g., vegetation and crop type, soil type, and minerals). Fine spectral information of HS data also helps in crop or vegetation condition assessment, disease detection, biophysical variables estimation, and water quality monitoring. However, spatial and temporal coverages of the Earth surface using HS remote sensing sensors (especially space-borne sensors) are very limited [1]. The tradeoff between spectral and spatial resolution of optical sensors causes coarse spatial resolution and narrow swath of large number of HS bands [2]. For example, Earth Observing 1 (EO-1) satellite carried both MS [i.e., Advanced Land Imager (ALI)] and HS (i.e., Hyperion) sensors but ALI products have swath of 37 km, whereas Hyperion products have a swath of only 7.5 km with the same spatial resolution (i.e., 30 m). Acquisitions of HS data are comparatively difficult and expensive. On the other hand, MS remote sensing data are readily available and cheaper, and also providing continuous spatial and temporal coverage of the Earth's surface since a long time using the spectral bands of coarser resolution. Therefore, development of MS to quasi-HS data transformation techniques can be investigated extensively in order to provide quasi-HS information in the absence of original one and to overcome the tradeoffs of remote sensing images and provide spectrally, spatially, and temporally improved synthetic images.

Numerous studies investigated the construction of MS bands by aggregating field spectra or HS bands [3]. Conversely, few attempts were made to transform MS data into HS data, assuming that vice-versa relation holds good. Chen *et al.* [4] simulated HS images from low spectral resolution (i.e., MS) images and demonstrated availability of additional information in the simulated images by generating classification images.

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Liu *et al.* [3] simulated HS bands (i.e., EO-1 Hyperion bands) from MS ALI data using spectral reconstruction approach and evaluated the simulated data by visual interpretation, statistical comparison, and classification. Spectral reconstruction approach was employed for simulation of EO-1 Hyperion data from EO-1 ALI imagery [5] and Landsat 8 Operational Land Imager (OLI) imagery [6] with application to land cover classification. A spectral resolution enhancement method (SREM) was proposed to generate wide swath HS images from remotely sensed MS images using auxiliary MS/HS data [2]. Hoang and Koike [7] developed a method, pseudo-HS image synthesis algorithm (PHISA), to transform Landsat imagery into quasi-HS imagery, which was extended to simulate quasi-HS imagery from EO-1 ALI imagery in [8]. Hoang and Koike [1] extended the PHISA method and developed pseudo-HS image transformation algorithm (PHITA) for transforming Landsat 7 Enhanced Thematic Mapper Plus (ETM+) imagery into quasi-Hyperion imagery with application to metal deposit-related minerals identification and mapping. Hoang and Koike [9] continued the use of PHITA to transform Landsat Thematic Mapper (TM), OLI, and EO-1 ALI sensor images into quasi-Hyperion images with application to minerals detection. PHITA does not consider all the MS bands to develop regression relationship with each HS band; instead, it uses Bayesian model averaging (BMA) approach for selection of optimal combinations of MS bands. Though PHITA was proven to be effective in MS to quasi-HS data transformation with application to minerals identification and mapping, this algorithm considers only linear relationship between optimal combination of MS bands and each HS band. All the HS bands may not hold the linear relationship with the MS bands. Hence, nonlinear relationship between MS and HS bands should be explored for better construction of quasi-HS data.

Deep neural networks or convolutional neural networks (CNNs) were employed successfully for construction of super-resolution (i.e., high-spatial resolution) images [10], [11] or satellite images [12] from low-spatial resolution images. Similarly, deep learning-based techniques can be explored for fine spectral resolution image creation. Therefore, we are proposing the use of deep learning-based convolutional neural network regression (CNNR) in this work to transform MS data into quasi-HS data by developing nonlinear relationship. CNNR was not implemented for MS to quasi-HS image transformation till date. CNNR has the additional advantages of assimilation of spatial information and utilization of deep learning-based spectral–spatial features in the regression model which is not the case with simple linear and nonlinear regression models. Deep learning techniques have shown definitive classification performances with high-dimensional remote sensing images [13]–[15]. CNN architecture uses local connections and shared weights for parameter estimation which reduces computational cost and extracts spectral–spatial features from the data, whereas other deep learning techniques, viz., deep belief network (DBN), stacked auto-encoder (SAE) extract spectral features involving estimation of too many parameters [16], [17]. Several studies proved the efficacy of deep learning technique, CNN for classification of MS and HS images [18]–[22]. The advantages of CNN algorithm (i.e., high level spectral–spatial feature representation, use of shared weights) were utilized in few studies [13], [23]–[25] to solve regression problems.

The main novelties and contributions of this article are to develop the nonlinear relationship between MS and HS bands using deep learning-based CNNR algorithm and transformation of MS to quasi-HS data. The quality of the CNNR-based quasi-HS data is compared with original HS data and evaluated in terms of different statistical measures. Stepwise linear regression (SLR) and support vector regression (SVR) are used as simple linear and nonlinear techniques, respectively, for creation of quasi-HS data and compared with CNNR-based product. The existing PHITA algorithm is also implemented in this study for creation of quasi-HS data to compare with the performances of proposed CNNR-based approach. Later, the developed models are used in a different location over an Indian scene, where original HS data are not available, to transform MS data into quasi-HS data and evaluated with application to crop classification. These models are also evaluated by comparing the original and quasi-HS crop spectra over a United States scene.

The remainder of this article is organized as follows. Section II presents the details about the used datasets and study area. Section III describes the proposed methodology and relevant techniques. The results of all the computational experiments are reported and discussed in Section IV. Section V summarizes the overall conclusions of this work.

II. Datasets

A. Satellite Data

1) Indian Scene: We have used Landsat 7 ETM+ and EO-1 Hyperion image scenes (Table I) as MS and HS data, respectively, partially covering the Karnataka state of India, for model building and testing by evaluating the quality of the quasi-HS data. Both satellite products are downloaded from https://earthexplorer.usgs.gov/. On-demand Level-2 surface reflectance products of Landsat 7 ETM+, which were radiometrically calibrated and atmospherically corrected by Landsat ecosystem disturbance adaptive processing system (LEDAPS), are downloaded. Focal analysis is performed with 3×3 mean filter for gap filling of Landsat 7 ETM+ images. Radiometric and atmospheric correction of Hyperion data is performed using fast line-of-sight atmospheric analysis of hypercubes (FLAASH) tool in ENVI software to generate the surface reflectance product. In case of Hyperion data, after removal of uncalibrated, noisy, and water-absorption bands, only 170 spectral bands among 242 bands are retained which are 8–57 (0.427–0.885 μm), 79–120 (0.996–1.346 μm), 131–165 (1.457–1.800 μm), and 181–223 (1.962–2.385 μm), and these 170 bands are utilized for all the experiments. The standard false color composite (STD FCC) images of Landsat 7 and Hyperion scenes, which are utilized for model building and quasi-HS data quality evaluation through statistical measures, are shown in Fig. 1. The common part of both the scenes is extracted and co-registered, and after removal of cloudy pixels, rest of the pixels are utilized in this analysis.

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TABLE I
SPECIFICATIONS OF SATELLITE DATA USED IN THIS STUDY

Satellite	Sensor	Acquisition date	Path/Row	Spatial resolution	Application
Landsat 7	ETM+	27/06/2014	144/52	30 m	Model building
EO-1	Hyperion	27/06/2014	144/52		Model building and testing
Landsat 7	ETM+	21/12/2015	146/48,49, 50,51		Simulation of quasi-HS data and application to crop classification
		23/12/2015	144/50,51,52		
		30/12/2015	145/47,48,49, 50,51,52		
Landsat 8	OLI	22/09/2016	29/38		Model building
EO-1	Hyperion	19/09/2016	29/38		Model building and testing, and Spectra comparison between original and quasi-HS data

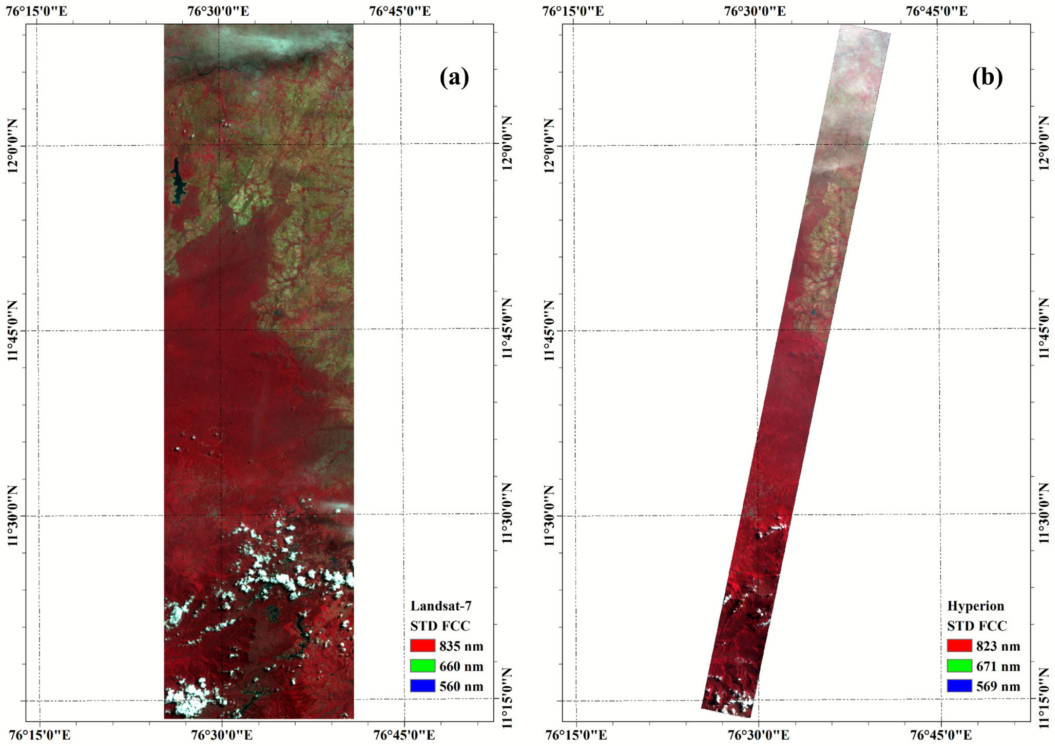


Fig. 1. STD FCC images of the area of interest derived from (a) Landsat 7 ETM+ and (b) Hyperion data.

MODIS land cover type 1 product (i.e., IGBP classification scheme of MCD12Q1) [Fig. 2(a)] is also used for this area to evaluate the simulated quasi-HS data for different land cover types. The land cover type data are downscaled to 30-m spatial resolution and the details of number of pixels corresponding to each type are presented in Table II. It is observed that maximum part of the study area (∼85.6%) falls under the land cover type "Croplands". This area partially covers two districts (i.e., Mysore and Chamarajanagar) of Karnataka and experiences semiarid climate and belongs to southern dry agro-climatic zone of Karnataka.

Landsat 7 ETM+ images of December 2015 over the ground-truth locations are utilized for crop classification application. These MS images are transformed into quasi-Hyperion data using the developed model and later this quasi-HS data are utilized to classify the crop types.

2) United States Scene: We have also used one Landsat 8 OLI and one EO-1 Hyperion image scenes (Table I) as MS and HS data, respectively, which partially cover some part of the five counties (viz., Mitchell, Nolan, Coke, Tom Green, and Irion) of the Texas state in the south-central region of the United States (Fig. 3), for model evaluation by comparing the crop spectra from original and quasi-HS data. These five counties are part of Central Texas and mostly covered by hills and rivers, and experience partially semiarid and subtropical climate. On-demand Level-2 surface reflectance products of Landsat 8 OLI data, which were radiometrically calibrated and atmospherically corrected by Landsat 8 surface reflectance code (LaSRC), are downloaded from https://earthexplorer.usgs.gov/. STD FCC image of the Hyperion data is overlapped over STD FCC image of Landsat 8 for this scene and presented in Fig. 3. These two images

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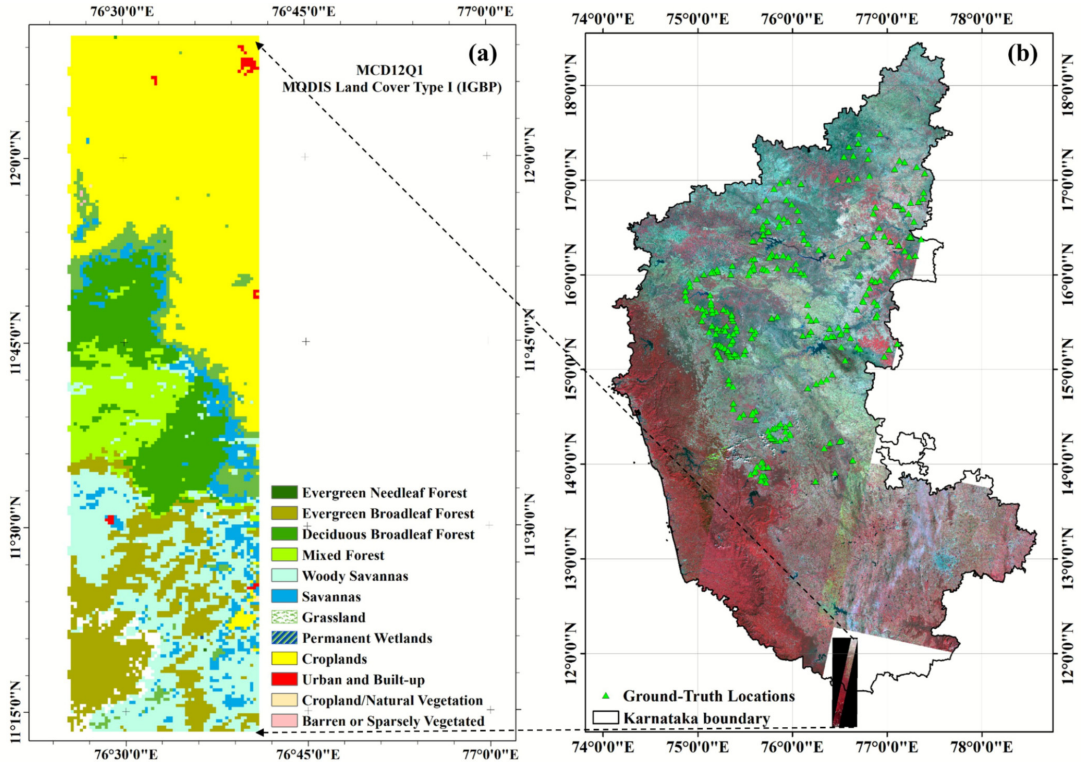


Fig. 2. (a) MODIS land cover map of the area of interest. (b) Locations of the area of interest and ground-truth crop information over Karnataka state.

TABLE II
DETAILS OF THE NUMBERS OF PIXELS FOR EACH LAND COVER TYPE (DERIVED FROM MODIS LAND COVER PRODUCT) OVER THE COMMON LANDSAT 7 AND HYPERION SCENE OF THE KARNATAKA STATE

Land cover type	Training pixels	Testing pixels
Evergreen broadleaf forest	2,968	2,918
Deciduous broadleaf forest	1,365	1,379
Mixed forest	1,966	1,930
Woody savannas	5,965	5,888
Savannas	1,873	1,936
Grasslands	6,277	6,092
Croplands	1,47,265	1,47,547
Urban and built-up	2,904	2,895

are utilized for model evaluation through statistical metrics and comparing the crop spectra between original and quasi-HS data. However, these two images do not have same acquisition dates and these are utilized here with the underlying assumption that the spectral signatures were almost the same. Both the images are co-registered and after removal of bad pixels, cloudy pixels, and pixels without crop class, rest of the pixels are utilized for the analysis.

B. Crop Data

1) Indian Scene: The crop information (i.e., ground truth) was provided by the Mahalanobis National Crop Forecast Centre (MNCFC), New Delhi, India, and was utilized in [26] and [27] for classification of crops. The locations of the acquired crop information are presented through green color dots in Fig. 2(b). This dataset has ground-truth information of eight types of Rabi crops over the 14 districts of Karnataka state of India. These districts experience both arid and semiarid climate and belong to six different agro-climatic zones (viz., North-Eastern Transition, North-Eastern Dry, Northern Dry, Central Dry, Southern Transition, and Northern Transition Zone) of Karnataka. The detailed information about the crops and number of ground-truth samples per crop type are provided in Table III. Average spectral reflectance curves of these eight types of crops are derived from the Landsat 7 ETM+ data and plotted in Fig. 4. The average spectral reflectance patterns are not clearly distinguishable in the visible range, whereas in the near infrared and short-wave infrared ranges, almost distinct spectral responses are observable.

2) United States Scene: The United States Department of Agriculture (USDA), National Agricultural Statistics Service (NASS) Cropland Data Layer (CDL), which provides crop-specific land cover types at 30-m spatial resolution, is downloaded for the year 2016 from https://nassgeodata.gmu.edu/CropScape/ and utilized here as reference ground-truth information. Four crop classes are identified in the common

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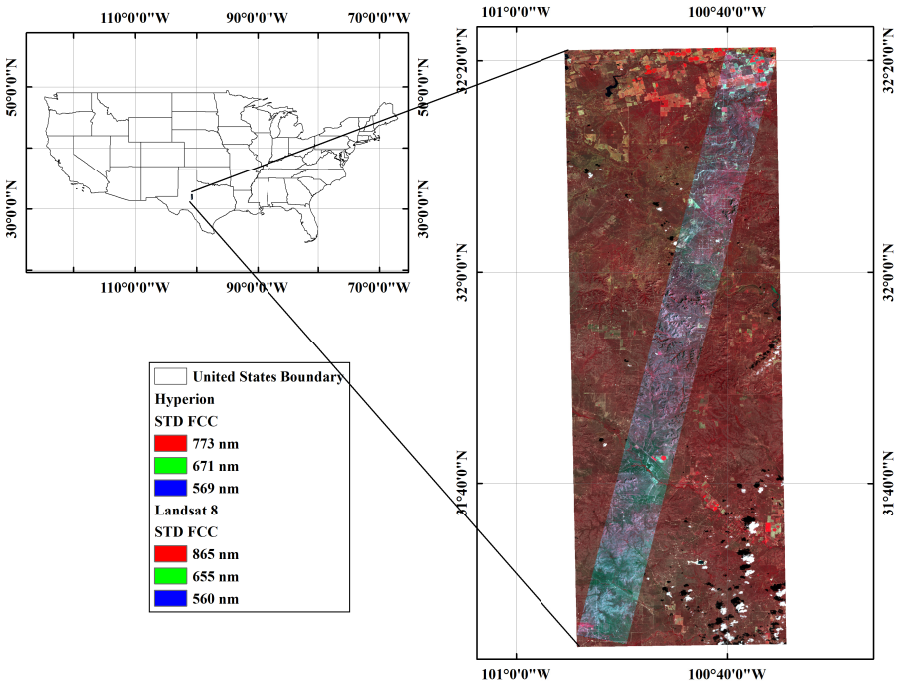


Fig. 3. Location of the area of interest over the United States and STD FCC image of Hyperion overlapped over Landsat 8 image scene.

TABLE III
DETAILS OF THE GROUND-TRUTH INFORMATION (i.e., CROP TYPES AND THE NUMBERS OF PIXELS PER CROP)

Sl No	Class Name	Number of Pixels
1	Groundnut	12
2	Cotton	35
3	Paddy	9
4	Jower	467
5	Bengalgram	231
6	Chilli	240
7	Toor	69
8	Wheat	23
Total Pixels		**1,086**

TABLE IV
DETAILS OF CROP CLASS INFORMATION FROM USDA/NASS CDL DATASET

Sl No	Class Name	Number of Pixels
1	Cotton	229
2	Sorghum	13
3	Winter wheat	80
4	Fallow cropland	27

Landsat 8 and Hyperion scene, which are listed in Table IV with corresponding number of pixels. Average spectral reflectance curves of these classes, derived from the Hyperion data, are presented in Fig. 5. These crop classes do not have significant differences in their crop spectra; hence, these are not utilized for crop classification application.

III. METHODS

The proposed methodology for MS to quasi-HS data transformation (with an example of Landsat 7 as MS and Hyperion as HS data) and evaluation of the quality of the quasi-HS data are presented briefly through a flowchart in Fig. 6. CNNR algorithm is proposed for developing the nonlinear relationship between MS and HS data and transformation of MS data to quasi-HS data utilizing the deep learning-based spectral–spatial features. MS to quasi-HS transformation is also carried out using PHITA, SLR, and SVR techniques to compare the performance of CNNR model. The existing PHITA model uses BMA approach for the selection of optimal combination of MS bands corresponding to each HS band and consecutively uses the optimal MS bands in simple multiple linear regression (MLR) for modeling. SLR is a simple and systematic method for adding and removing predictor terms or variables (i.e., MS bands) from a linear or generalized linear model based on their statistical significance in explaining the response variable. PHITA and SLR are implemented here as simple linear models. SVR and CNNR are the nonlinear models, where Gaussian kernel as a kernel function and rectified linear unit (relu) as an activation function are utilized, respectively, to deal with the nonlinearities in the regression modeling. PHITA and SLR may or may not use all the six Landsat 7 ETM+ or seven Landsat 8 OLI MS bands as predictor variables; however, SVR and CNNR use all the six or seven bands to develop the relationship with each of the 170 Hyperion bands. CNNR considers spatial information and develops deep learning-based nonlinear relationship,

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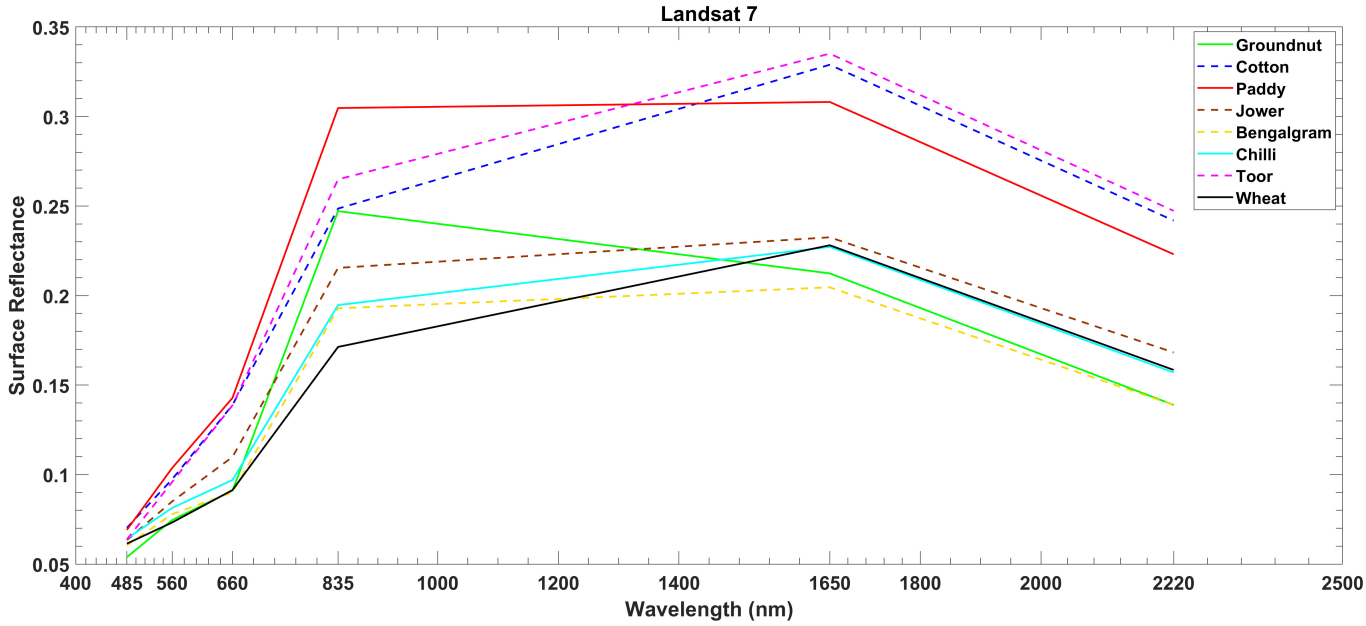


Fig. 4. Average spectral reflectance curves (derived from Landsat 7 ETM+ data) of the eight types of crops.

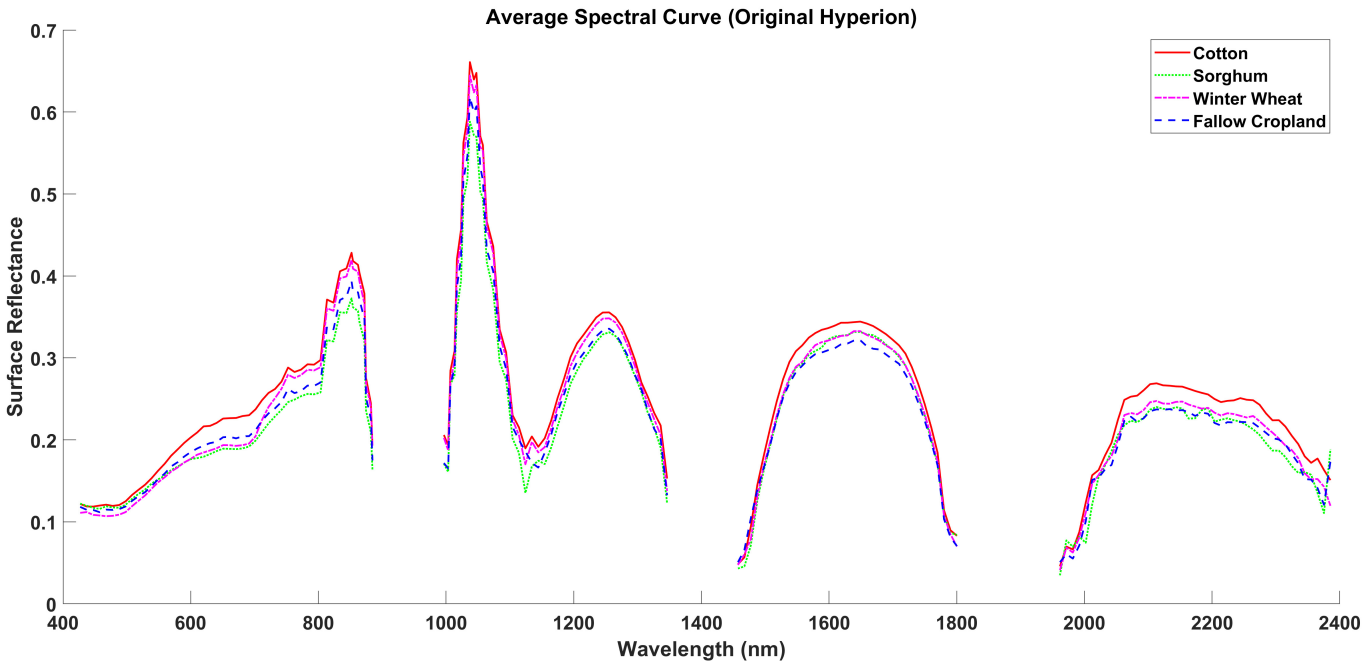


Fig. 5. Average spectral reflectance curves (derived from Hyperion data) of the four types of crop classes of USDA/NASS CDL data.

which can be advantageous compared to simple linear (i.e., PHITA and SLR) and nonlinear (i.e., SVR) techniques. Therefore, to prove the advantages of CNNR-based MS to quasi-HS data transformation, it is compared with the PHITA, SLR, and SVR models. Quasi-HS data are compared with original HS data for different land cover types using numerous statistical measures. The developed regression models are also evaluated for a different location over the Karnataka state (i.e., Indian scene) with application to crop classification, where Landsat 7 ETM+ data are transformed into quasi-Hyperion data using the developed models and utilized to classify the crops. The crop classification application is carried out for such an area where original HS images are not available. Therefore, efficient performance of quasi-HS data in the absence of original HS data will generalize the effectivity and wide applicability of the proposed regression model. Similarly, another application is carried out for a United States scene to transform Landsat 8 data into Hyperion data, and later simulated quasi-Hyperion data derived crop class spectra are compared with the original Hyperion spectra.

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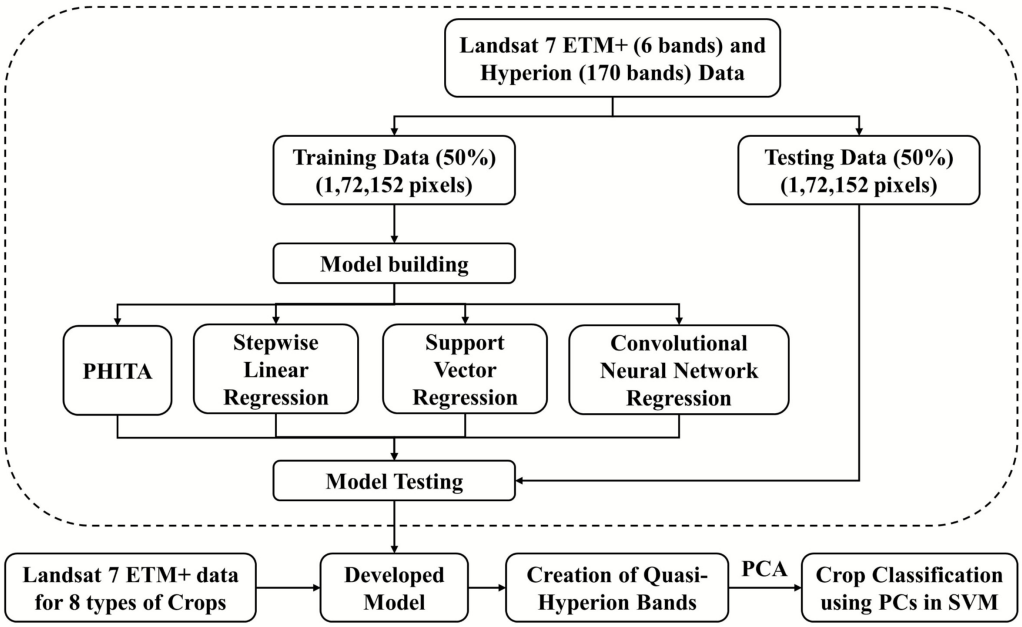


Fig. 6. Workflow of the proposed methodology.

TABLE V
DETAILS OF THE USED CNNR MODELS

Models	Details of CNNR Model's Network Architecture
CNNR_M1	Input (5×5×6) → CNN Layer (Filters: 32→16→8) → Flatten → NN Layer (170 Units, Dense) → Dropout (0.5) → NN Layer (170 Units, Dense) [Output]
CNNR_M2	Input (5×5×6) → CNN Layer (Filters: 16→8) → Flatten → NN Layer (170 Units, Dense) → Dropout (0.5) → NN Layer (170 Units, Dense) [Output]
CNNR_M3	Input (5×5×6) → CNN Layer (Filters: 16→8) → Flatten → NN Layer (100 Units, Dense) → Dropout (0.5) → NN Layer (170 Units, Dense) [Output]
CNNR_M4	Input (5×5×6) → CNN Layer (Filters: 16→8→4) → Flatten → NN Layer (50 Units, Dense) → Dropout (0.5) → NN Layer (170 Units, Dense) [Output]
CNNR_M5	Input (5×5×6) → CNN Layer (Filters: 16→8) → Flatten → Dropout (0.5) → NN Layer (170 Units, Dense) [Output]
CNNR_M6	Input (5×5×6) → CNN Layer (Filters: 16→32→64) → Flatten → NN Layer (170 Units, Dense) → Dropout (0.5) → NN Layer (170 Units, Dense) [Output]

A. Convolutional Neural Network Regression

CNN is a deep learning-based technique, which uses multilayer neural networks (NNs) to learn deep spectral–spatial features from the data. CNN architecture uses "convolution" operation in the hidden layers, which introduces sparse interactions, parameter sharing, and equivariant representations in the network and helps in reducing memory requirements when compared with simple multilayer NN architecture [28]. In the proposed CNNR model, narrow spectral bands of Hyperion are the predictand variables and broadbands of Landsat 7 ETM+ are considered as the predictor variables, which can be explained using the following equation:

$$Y_i = f(X_1, X_2, X_3, X_4, X_5, X_7) \tag{1}$$

where Y_i is the reflectance of ith band of Hyperion; X_1, X_2, X_3, X_4, X_5, and X_7 are the reflectances of bands 1, 2, 3, 4, 5, and 7, respectively, of Landsat 7 ETM+; $f(.)$ is a nonlinear transformation function. In case of Landsat 8 to Hyperion transformation, seven MS bands are considered as predictor variables.

Six different CNNR architectures, which are briefly tabulated in Table V, are investigated by changing the numbers of the convolution and NN layers, and sizes of filters and hidden units. Since there are no general rules for choosing the sizes of filters and convolutional layers, optimal combination (or size) is chosen through trial-and-error approach by considering different combinations [29]. The number of convolutional layers is limited to three layers here, so that the CNNR architectures do not become computationally complex and expensive. Sizes of the filters in each convolution layer are varied in both decreasing and increasing order to identify the optimal performing architecture. All these architectures are presented by considering Landsat 7 as MS data. Similarly, these can be extended with Landsat 8 as MS data, where the number of predictor variables increases to seven from six. First, CNNR (i.e., CNNR_M1) architecture of Table V is explained in detail through a schematic in Fig. 7. In the

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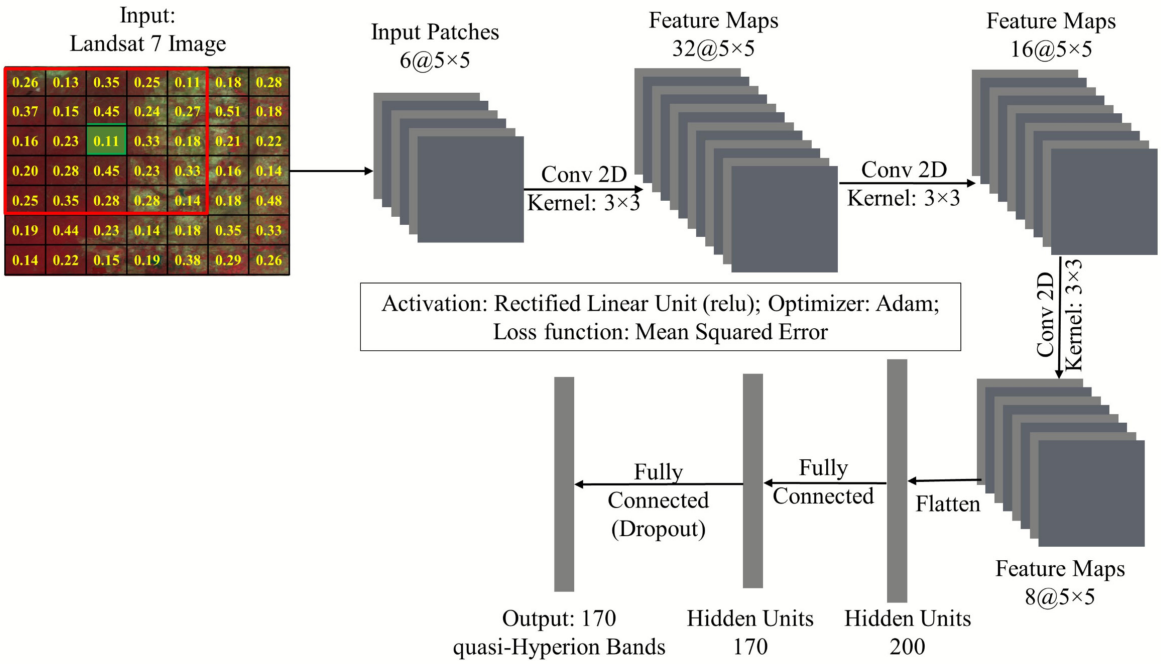


Fig. 7. Architecture of the proposed CNN-based regression technique.

first step, small patches of (5 × 5) size are extracted corresponding to each pixel of the Landsat 7 ETM+ image by centering the pixel in the patch, as highlighted in the "Input: Landsat 7 Image" part of Fig. 7. Then, 2-D convolutional filters with a kernel size of (3 × 3) are operated on the image patches and feature maps consecutively. This architecture uses filter size of 32, 16, and 8 sequentially. The sizes of the filters and numbers of convolutional layers are changed to identify better architecture in terms of performance evaluation. Later, the feature maps of last convolution layer are flattened and connected with different NN layers to generate the reflectances of quasi-Hyperion bands. Similar to the convolutional layers, NN layers and numbers of hidden units are varied. All the CNNR models are using "relu" as activation function, "adam" as optimizer, and "mean squared error" (mse) as loss function. Keras library [30] of Python programming language is used to implement the CNNR architecture.

B. Model Testing Methods

Quality of the quasi-Hyperion data is evaluated using different statistical metrics, which can be categorized into two classes, viz., band-wise and pixel-wise evaluation. Quasi-Hyperion data are evaluated with only testing samples. Four statistical metrics, viz., Pearson's correlation coefficient (PCC), root-mean-square error (RMSE), peak signal-to-noise ratio (PSNR), and structural similarity (SSIM) index [1], [9] are used for band-wise evaluation. Average statistics of all the bands are shown via boxplots. Spectral angle mapper (SAM), spectral information divergence (SID), and RMSE are used for pixel-wise evaluation of quasi-Hyperion data. SAM, SID, and RMSE_j compare the spectral reflectance curves between original and quasi-Hyperion images corresponding to each pixel. Thereafter, average statistics of all the pixels are presented through boxplots. All these statistical metrics are calculated using (2)–(8) in MATLAB. It is important to calculate these different types statistical metrics since each metric is not self-sufficient to unveil all characteristics of simulated quasi-HS image quality [9].

We have performed a crop classification application using the quasi-Hyperion data to evaluate the efficiency of the regression models and quality of the generated data. In case of crop classification, overall accuracy (OA), kappa coefficient (k), average accuracy (AA), user accuracy (UA), and producer accuracy (PA) are used as a performance evaluation measures.

PCC measures the linear correlation between the original and quasi-Hyperion bands using the following equation:

$$\mathrm{PCC}_i = \frac{\sum_{j=1}^{n}\left(\rho_{ij}^{\mathrm{H}}(\lambda)-\mu_i^{\mathrm{H}}\right)\left(\rho_{ij}^{\hat{H}}(\lambda)-\mu_i^{\hat{H}}\right)}{\sqrt{\sum_{j=1}^{n}\left(\rho_{ij}^{\mathrm{H}}(\lambda)-\mu_i^{\mathrm{H}}\right)^2}\sqrt{\sum_{j=1}^{n}\left(\rho_{ij}^{\hat{H}}(\lambda)-\mu_i^{\hat{H}}\right)^2}} \tag{2}$$

where H and $\hat{H}$ represent the original and quasi-Hyperion images, respectively; $\rho_{ij}^{\hat{H}}(\lambda)$ is reflectance of quasi-Hyperion image and $\rho_{ij}^{\mathrm{H}}(\lambda)$ is reflectance of original Hyperion image at co-registered pixel j and band i having wavelength of λ; μ_i^{H} and $\mu_i^{\hat{H}}$ are means of all pixel values in band i; and n is the total number of pixels in band i.

RMSE measures the difference between refelectances of quasi-Hyperion and original Hyperion images using the following equation:

$$\mathrm{RMSE}_i = \frac{\sum_{j=1}^{n}\sqrt{\left(\rho_{ij}^{\hat{H}}(\lambda)-\rho_{ij}^{\mathrm{H}}(\lambda)\right)^2}}{n}. \tag{3}$$

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In case of pixel-wise evaluation also, RMSE can be used by modifying the equation as follows:

$$\mathrm{RMSE}_j = \frac{\sum_{i=1}^{N} \sqrt{\left(\rho_{ij}^{\hat{H}}(\lambda) - \rho_{ij}^{\mathrm{H}}(\lambda)\right)^2}}{N} \tag{4}$$

where N is the number of HS bands.

PSNR is a widely used statistical metric for quality evaluation of reconstructed image (i.e., quasi-HS) [9], which can be calculated using the following equation:

$$\mathrm{PSNR} = 10\log_{10}\left(\mathrm{peakval}^2/\mathrm{MSE}\right) \tag{5}$$

where peakval is taken from the range of the image data type (e.g., for unsigned integer (uint) 8-bit image, peakval = 255). MSE is the mean square error.

SSIM is an image quality metric that assesses the visual impact of three characteristics of an image, viz., luminance, contrast, and structure into a single local quality score [31]

$$\mathrm{SSIM} = \frac{\left(2\mu_{\mathrm{H}}\mu_{\hat{H}} + C_1\right)\left(2\sigma_{\mathrm{H}\hat{H}} + C_2\right)}{\left(\mu_{\mathrm{H}}^2 + \mu_{\hat{H}}^2 + C_1\right)\left(\sigma_{\mathrm{H}}^2 + \sigma_{\hat{H}}^2 + C_2\right)} \tag{6}$$

where μ and σ are the mean and variance of the pixel reflectances, respectively, and C_1 and C_2 are constants used to stabilize the calculation when $\mu_{\mathrm{H}}^2 + \mu_{\hat{H}}^2$ or $\sigma_{\mathrm{H}}^2 + \sigma_{\hat{H}}^2$ is close to zero. MSSIM is the mean SSIM of all bands; perfect agreement is reflected by MSSIM = 1.

SAM evaluates the difference between two spectra by measuring the angle (α)

$$\alpha_j = \cos^{-1}\frac{\sum_{i=1}^{N}\rho_{ij}^{\hat{H}}(\lambda)\rho_{ij}^{\mathrm{H}}(\lambda)}{\sqrt{\sum_{i=1}^{N}\rho_{ij}^{\hat{H}}(\lambda)^2}\sqrt{\sum_{i=1}^{N}\rho_{ij}^{\mathrm{H}}(\lambda)^2}}. \tag{7}$$

SID is an information theoretic spectral metric. SID considers each pixel as a random variable and uses its spectral histogram to define a probability distribution. The spectral similarity between two pixels is measured by the discrepancy of probabilistic behaviors between their spectra using the following equation:

$$\mathrm{SID}_j = D\left(\rho_{ij}^{\mathrm{H}}(\lambda) \parallel \rho_{ij}^{\hat{H}}(\lambda)\right) + D\left(\rho_{ij}^{\hat{H}}(\lambda) \parallel \rho_{ij}^{\mathrm{H}}(\lambda)\right) \tag{8}$$

where $D(\rho_{ij}^{\mathrm{H}}(\lambda) \parallel \rho_{ij}^{\hat{H}}(\lambda)) = \sum_{i=1}^{N} p_i \log(p_i/q_i)$; $D(\rho_{ij}^{\hat{H}}(\lambda) \parallel \rho_{ij}^{\mathrm{H}}(\lambda)) = \sum_{i=1}^{N} q_i \log(q_i/p_i)$; and $p_i = r_j/\sum_{i=1}^{N} r_i$; $q_i = \rho_{ij}^{\hat{H}}(\lambda)/\sum_{i=1}^{N}\rho_{ij}^{\hat{H}}(\lambda)$.

C. Computational Experiment Setup

All the computational experiments are carried out in a desktop with Intel Core i5-4460 CPU at 3.20-GHz processor, 16.00-GB memory (RAM), and NVIDIA GeForce 210 graphics card.

To build the regression models, 50% of the pixels from the common area of Landsat 7 ETM+ and Hyperion images over the Indian scene are randomly chosen and rest of the 50% pixels are used to test the models. Deep learning-based CNNR models generally require sufficient samples to avoid the issue of overfitting; hence, the whole dataset is divided equally into two parts. Similarly, over the United States scene, Landsat 8 and Hyperion data are divided into two parts, viz., 50% as training and 50% as testing samples. The statistical measures for evaluation of the quasi-Hyperion data are calculated only with the testing samples. In case of PHITA, BMA is performed using "BMA" package [32] in R software and MLR is performed in MATLAB. SLR and SVR are performed in the MATLAB environment, whereas experiments with CNNR models are carried out in Python environment using the Python-based deep learning library "Keras" [30].

In order to evaluate the quasi-Hyperion data in crop classification application over the Indian scene, experiments are carried out in the MATLAB environment. Support vector machine (SVM) classifier with Gaussian kernel has been used for classification of the crops by considering 50% of the pixels from each crop class as training samples and rest of the pixels as testing samples. This training and testing samples partitions are done randomly and such ten random trials are performed. Mean ± standard deviation (SD) of OA, k, and AA from these ten trials are reported for each classification experiment. UA and PA of each crop type are also reported.

IV. Results and Discussion

A. Band-Wise Evaluation of Quasi-Hyperion Data Over the Indian Scene

The quasi-Hyperion data are generated and evaluated with nine models, viz., PHITA, SLR, SVR, and six CNNR models considering different architectures. The statistical measures (i.e., PCC, RMSE, PSNR, and SSIM) are calculated for each narrow spectral band comparing the reflectance values of original and quasi-Hyperion bands for all the testing pixels. PCC and SSIM close to 1, lower the values of RMSE, and higher the values of PSNR signify better simulation of reflectance values. The statistical measures are calculated for all the 170 spectral bands and presented through a boxplot for each model (Fig. 8). Here, the outliers indicate that few specific spectral bands are performing differently (i.e., either poorly or better) compared with the most of the bands, and the values of the statistical measures are calculated to be beyond 1.5 times of difference between 25th (q_1) and 75th (q_3) percentiles [i.e., maximum whisker length × ($q_3 - q_1$)] from the 25th and 75th percentiles. In case of PCC, PSNR, and SSIM, outliers below 25th percentile and for RMSE outliers above 75th percentile represent poor performance. It is observed that PHITA and SLR are providing similar performances (Fig. 8). Though both PHITA and SLR use linear regression to develop the relationship and provide similar performances, SLR requires more computational time compared with PHITA for the selection of optimal combination of MS bands and to train the respective models (Table VI). It is evident from the boxplots of all the statistical metrics (Fig. 8) that all the CNNR models are able to create the quasi-Hyperion data more accurately compared with the PHITA, SLR, and SVR models. CNNR_M6 architecture, where three CNN layers are used with increasing order of filter size (i.e., 16, 32, and 64), is performing better when compared with all other CNNR models. However, CNNR_M6 uses maximum sizes of filters and requires more computational time to train the model compared to other CNNR architectures. CNNR_M2, CNNR_M3, and CNNR_M5 require minimum

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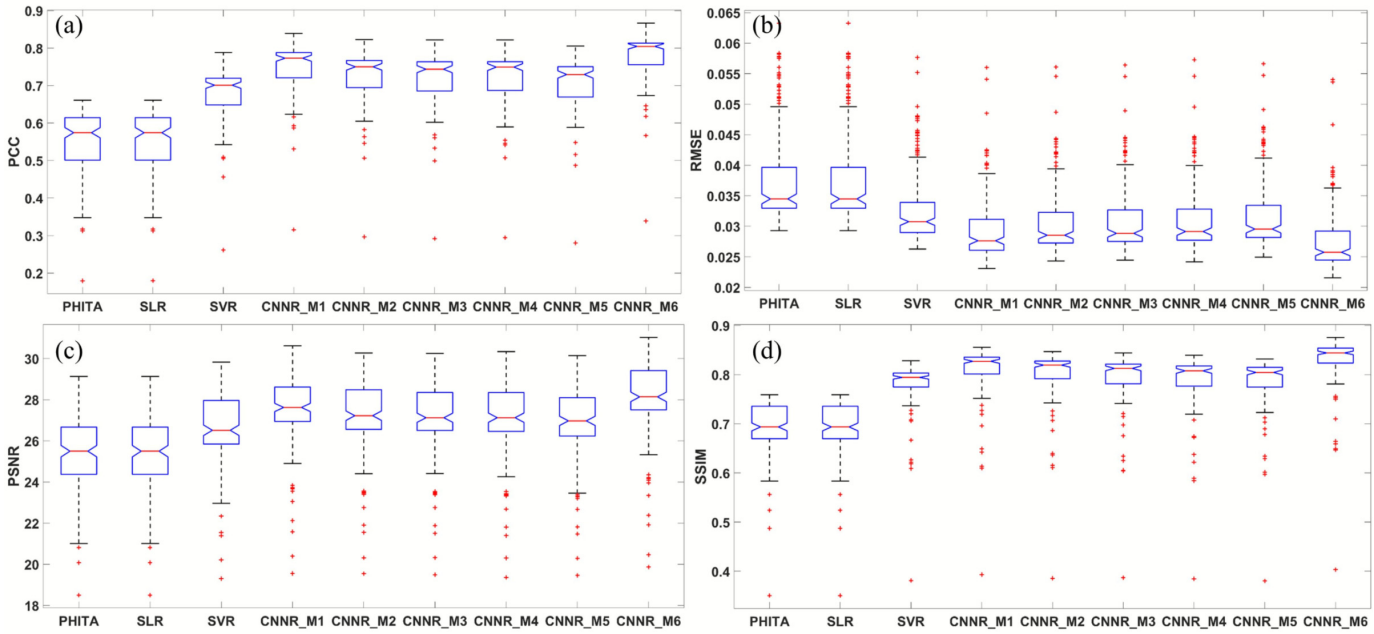


Fig. 8. Comparison of statistical measures (calculated between all the pixels of 170 original and quasi-Hyperion bands), (a) PCC, (b) RMSE, (c) PSNR, and (d) SSIM of different models.

TABLE VI
COMPUTATIONAL TIME REQUIRED TO TRAIN THE MODELS

Model	PHITA	SLR	SVR	CNNR_M1	CNNR_M2	CNNR_M3	CNNR_M4	CNNR_M5	CNNR_M6
Training time (min)	3.2	6.7	3,154	7.92	5	4.58	6.67	4.58	17.5

training time since only two convolutional layers are utilized. SLR provides most inferior performance which considers only linear relationship between the MS and HS bands. Therefore, the nonlinear relationship is more appropriate for developing the model and creating the quasi-Hyperion data. Though SVR provides better performance compared with linear models, PHITA and SLR, it needs high computational time for training compared to all other models. Furthermore, the CNNR models provide extra advantage of spatial convolution, which helps in developing better relationship and provides superior quasi-Hyperion data when compared with SVR. The average statistical metrics, viz., PCC, RMSE, PSNR, and SSIM improvise by 45.55, 26.6, 10.67, and 20.03%, respectively, with the use of CNNR_M6 as compared with the use of PHITA. In case of CNNR_M6, which is proven to be performing better compared with other models, eight spectral bands are identified to be outliers in all the statistical measures. These eight spectral bands are with serial numbers (wavelength): 94 (1467 nm), 128 (1962 nm), 129 (1972 nm), 132 (2002 nm), 133 (2012 nm), 168 (2365 nm), 169 (2375 nm), and 170 (2385 nm) among the 170 quasi-Hyperion bands. The outlier spectral bands belong to the near-infrared and mid-infrared wavelength region.

B. Band-Wise Evaluation of Quasi-Hyperion Data Over the Indian Scene for Different Land Cover Types

The quasi-Hyperion data are evaluated for different models considering the testing dataset of each land cover type separately. The information about the land cover type is adopted from MODIS land cover product (MCD12Q1). The evaluation of the statistical measures for all the eight land cover types is presented in Figs. 9 and S1–S7. Since PHITA is providing similar performance as SLR, land cover type-wise analysis is only presented with SLR. In case of all the land cover types, model CNNR_M6 is performing better (with higher median values and minimum interquartile range) in terms of all the statistical measures (i.e., PCC, RMSE, PSNR, and SSIM). It is observed that all the evaluated statistical measures are showing better performance in case of Croplands (Fig. 9), which are close to the general (i.e., irrespective of land cover classes) statistical evaluation. The prevailing performance of Croplands' pixels is evident because of its maximum coverage (i.e., 85.6%) of the study area. However, few other land cover types have performed exceptionally well with respect to a specific statistical measure. Deciduous broadleaf forest, Mixed forest, Croplands, and Urban and built-up land cover types are showing very high values of SSIM (i.e., median $\geq$ 0.85) (Figs. S2, S3, 9, and S7). Similarly, high values of PSNR (i.e., median $\geq$ 25) has been observed in case of Grasslands, Croplands, and Urban and built-up (Figs. S6, 9, and S7). Though in case of Urban and built-up (Fig. S7), PSNR and SSIM are evaluated to be very high, PCC is showing very average performance (i.e., median $\sim$ 0.68). The pixels belonging to Savannas class show very poor performance with reference to all the statistical measures (Fig. S5), which has only $\sim$1.1% coverage of the

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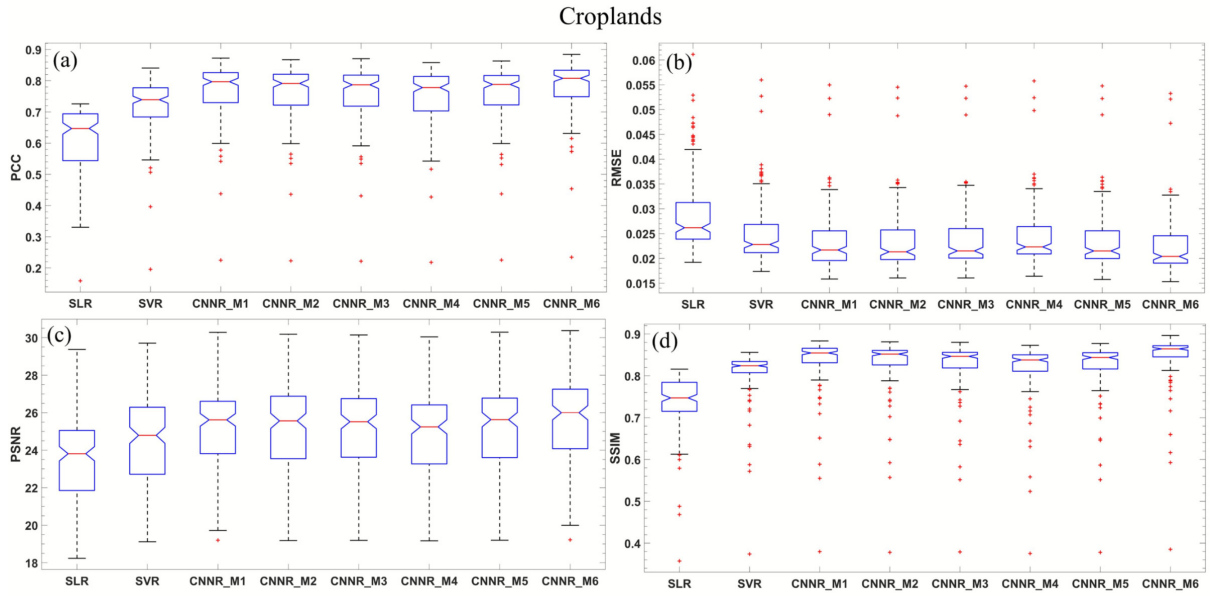


Fig. 9. Comparison of (a) PCC, (b) RMSE, (c) PSNR, and (d) SSIM calculated from different models considering testing data of land cover type, Croplands.

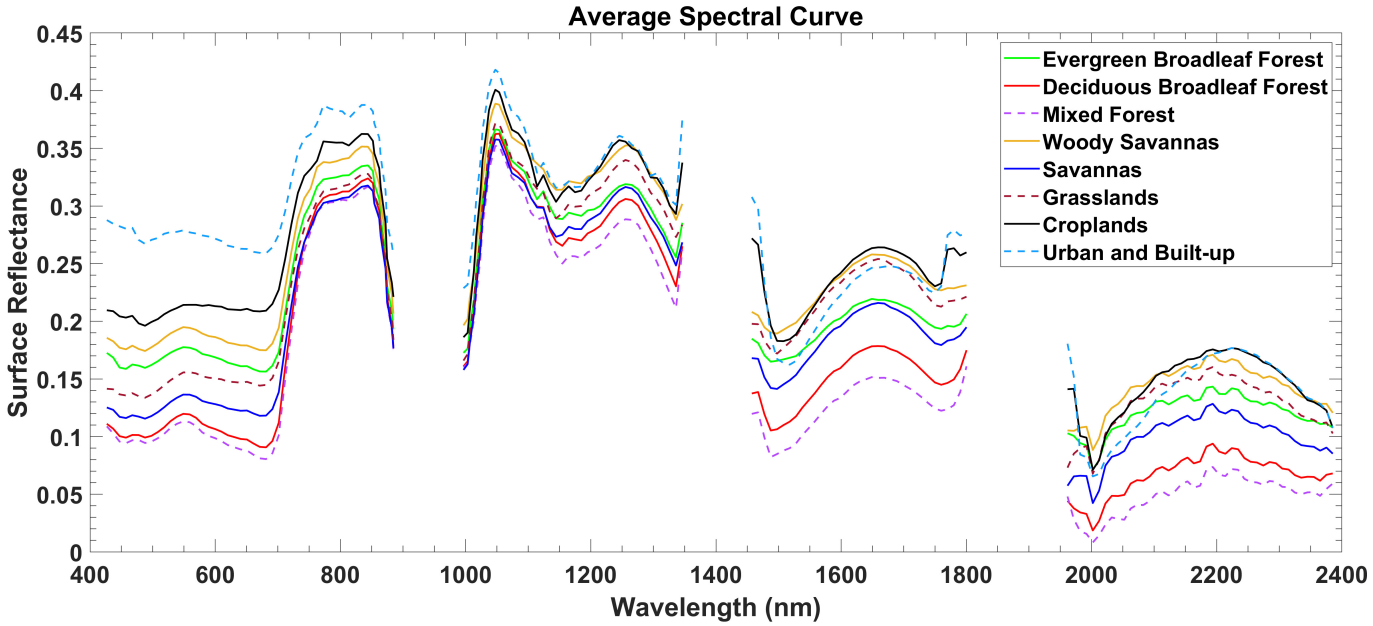


Fig. 10. Average spectral reflectance curves for all the land cover types derived from quasi-Hyperion data.

study area. The average spectral reflectance curves for all the land cover types are derived from the CNNR_M6 generated quasi-Hyperion data and plotted in Fig. 10. The average quasi-HS responses are distinguishable between different land cover types.

C. Pixel-Wise Evaluation of Quasi-Hyperion Data Over the Indian Scene for Different Land Cover Classes

In this section, simulated quasi-Hyperion data are evaluated by comparing the spectral reflectance curves of each pixel with the original Hyperion data. Statistical metrics SAM, SID, and RMSE are calculated to compare the spectral curves of each pixel. Since the land cover type Croplands has the maximum coverage in the study area and overall performance is mostly dominated by the performance of Croplands, here we have reported the evaluation performance only in case of Croplands (Fig. 11). It is evident from Fig. 11(a) and (c) that all the CNNR models are providing improved measures (i.e., SAM and RMSE) compared with SLR and SVR, whereas SID [Fig. 11(b)] shows almost equivalent range of values. SAM and RMSE are proven to be more sensitive in terms of presenting the improvements in performance. Since the number of pixels is very large (i.e., 1,47,547), the differences in performance among the models are not clearly distinguishable. In case of evaluation of statistical metrics with the model CNNR_M6, it is observed that measures of about 4.5% pixels are in the outlier range.

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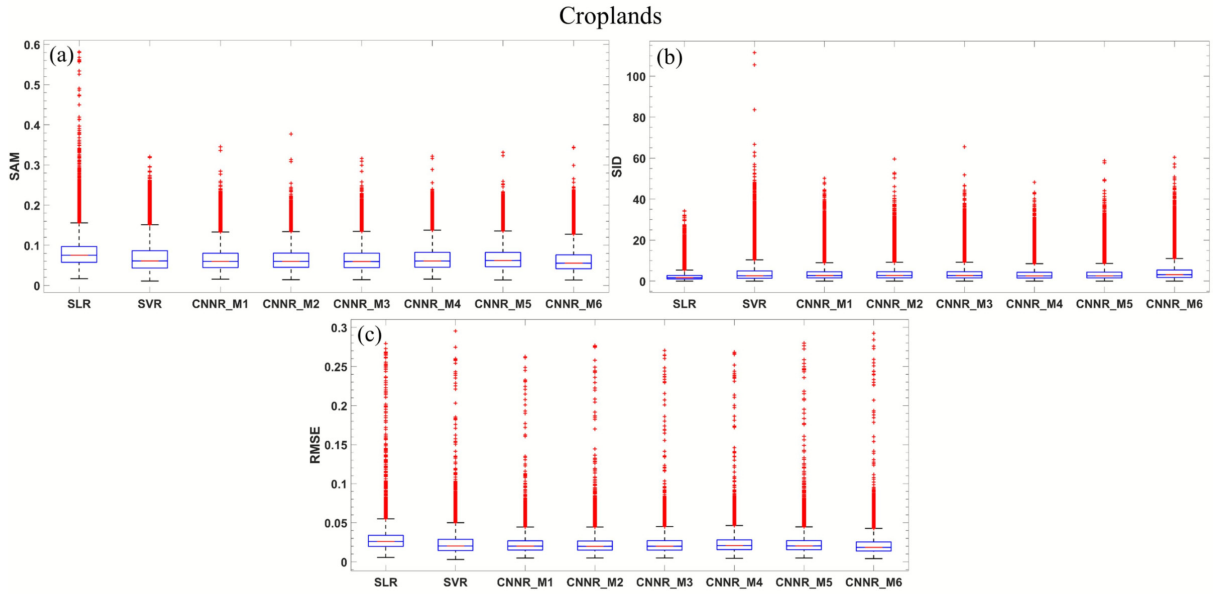


Fig. 11. Comparison of (a) SAM, (b) SID, and (c) RMSE calculated from different models considering testing data of land cover type, Croplands.

TABLE VII
CROP CLASSIFICATION PERFORMANCES WITH LANDSAT 7 ETM+ AND QUASI-HYPERION DATA

Model	Data	Features	OA (%)	k	AA (%)
	Landsat 7 ETM+	6 bands	77.94±1.50	0.6862±0.0219	73.72±5.91
PHITA	Quasi-Hyperion	170 Quasi-HS bands	73.39±2.00	0.6208±0.0312	63.73±5.79
SLR			73.11±1.47	0.6172±0.0205	62.21±5.96
PHITA		6 PCs (extracted from170 Quasi-HS bands)	76.76±1.84	0.6648±0.0298	76.37±6.75
SLR			76.12±3.21	0.6521±0.0566	76.54±6.34
SVR		10 PCs (extracted from170 Quasi-HS bands)	74.32±1.32	0.6333±0.0173	68.77±9.08
CNNR_M1			86.60±1.94	0.8100±0.0276	87.47±4.18
CNNR_M2			**88.36±1.52**	**0.8347±0.0219**	**88.87±4.11**
CNNR_M3			84.28±4.80	0.7714±0.0794	88.85±2.81
CNNR_M4			82.90±1.96	0.7568±0.0268	85.71±4.26
CNNR_M5			84.56±3.58	0.7777±0.0581	87.94±4.12
CNNR_M6			83.15±4.22	0.7587±0.0633	86.46±6.70

D. Evaluation of Quasi-Hyperion Data With Application to Crop Classification Over the Indian Scene

The pixels with eight types of crops are classified using the MS Landsat 7 ETM+ data and HS quasi-Hyperion data to compare the performances and evaluate the advantage of simulated quasi-Hyperion data. Classification performances of all the experiments are presented in Table VII. In case of quasi-Hyperion bands-based classification experiments, first 6 and 10 principal components (PCs) of 170 spectral bands are utilized since first 6 and 10 PCs explain most of the variability (i.e., ∼99%) in case of linear (i.e., PHITA and SLR) and nonlinear (i.e., SVR and CNNR) model-based quasi-HS data, respectively, and use of all the 170 bands does not provide better accuracy as experimented with PHITA and SLR-based quasi-Hyperion bands (Table VII). PHITA and SLR-based quasi-Hyperion bands perform similarly with crop classification application too. The inferior classification performance with the use of all the constructed quasi-HS bands is observed because of presence of redundancy in the HS data. It is observed that PHITA, SLR, and SVR-based

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TABLE VIII
USER AND PRODUCER ACCURACIES OF EACH CROP TYPE WITH DIFFERENT SETS OF FEATURES

	Model		SLR	SVR	CNNR_M2
	Features	6 bands of Landsat 7 ETM+	6 PCs of 170 quasi-HS bands	10 PCs of 170 quasi-HS bands	10 PCs of 170 quasi-HS bands
Groundnut	UA (%)	55.00±24.9	62.87±25.1	63.33±35.3	**71.67±30.0**
	PA (%)	30.00±15.3	33.33±17.6	25.00±14.2	**36.67±15.3**
Cotton	UA (%)	86.63±12.2	**93.91±6.34**	87.84±10.1	89.36±8.15
	PA (%)	75.00±12.9	67.22±14.7	68.89±11.8	**88.89±9.07**
Paddy	UA (%)	65.19±26.5	68.75±27.4	31.63±33.9	**100±0**
	PA (%)	47.50±21.9	32.50±29.0	25.00±33.3	**50.00±23.6**
Jower	UA (%)	80.95±2.06	77.54±6.06	78.27±1.93	**87.69±2.21**
	PA (%)	87.65±1.83	91.92±2.73	85.34±4.59	**93.80±2.39**
Bengalgram	UA (%)	75.31±3.63	75.62±6.48	72.64±4.23	**88.06±3.53**
	PA (%)	75.09±5.98	66.12±5.99	69.74±4.33	**90.17±3.90**
Chilli	UA (%)	76.63±3.19	74.60±2.81	69.99±3.17	**89.85±3.30**
	PA (%)	72.50±4.76	70.92±7.86	71.17±4.34	**85.92±5.01**
Toor	UA (%)	73.41±8.71	82.91±8.46	74.91±9.64	**93.03±3.90**
	PA (%)	61.47±9.65	49.12±17.6	54.12±11.7	**71.47±8.99**
Wheat	UA (%)	76.68±14.4	71.08±24.5	62.94±24.8	**91.31±11.8**
	PA (%)	55.83±9.66	42.50±11.4	41.67±11.1	**75.00±20.8**

quasi-Hyperion bands are not able to classify the crops as efficiently as Landsat 7 ETM+ bands. The general technical idea is that HS data should be more resourceful and should provide better classification accuracy when compared with the MS data but here we are observing contrary results with the constructed HS data. This implies that PHITA, SLR, and SVR-based regression models are not efficient enough to transform MS bands into resourceful quasi-HS bands. However, all the CNNR models-based quasi-Hyperion data are able to classify the crops with better accuracy when compared with the classification performance of Landsat 7 ETM+ bands. The poor performances of PHITA, SLR, and SVR-based quasi-Hyperion data are also noticeable with the evaluation of statistical metrics where CNNR models-based data are providing better results. The deep learning-based nonlinear regression approach, where spatial information is incorporated through "convolution" operation of CNN algorithm, is proven to be advantageous for MS to quasi-HS data transformation with application to crop classification too.

Here, CNNR_M2 model-based quasi-Hyperion data are providing best OA, k, and AA for classification of the crops. However, CNNR_M6 is performing better with reference to the statistical measures. In case of crop classification, CNNR_M6 generated bands are performing well with some trials and very poorly with other trials, as a result high SD of OA, k, and AA is observed compared with the performance of CNNR_M2-based bands. CNNR_M6 generated reflectance data of all the samples are not efficient enough to train the SVM classifier algorithm which is resulting in inconsistencies in the testing OA, k, and AA for different trials. The main difference between the CNNR_M2 and CNNR_M6 architectures is the number of convolutional layers, where the former one has two layers with filter sizes in decreasing order (i.e., 16 and 8) and the latter one has three layers with filter sizes in increasing order (i.e., 16, 32, and 64). Therefore, CNNR_M2 architecture is favorable in terms of computational complexity and time (Table VI). CNNR_M2 also performs reasonably well in terms of statistical metrics too and the average statistical metrics, viz., PCC, RMSE, PSNR, and SSIM are improved by 34.56, 18.34, 6.99, and 15.96%, respectively, with the use of CNNR_M2 as compared to the use of PHITA. From these computational experiment results, it can be concluded that CNNR_M2 model is the most generalized one which can be employed in a location where original HS data are not available.

UA and PA of each crop type are also calculated to evaluate the crop-wise classification performance and reported in Table VIII. UA and PA are calculated with four classification experiments considering different sets of features, viz: 1) six bands of Landsat 7 ETM+; 2) six PCs extracted from SLR-based 170 quasi-Hyperion bands. 3) ten PCs extracted from SVR-based 170 quasi-Hyperion bands; and 4) ten PCs extracted from CNNR_M2-based 170 quasi-Hyperion bands. Since CNNR_M2 is performing better in terms of overall classification performances compared with other CNNR-based models, UA and PA are calculated only for this model. It is evident that except UA of Cotton, all other UA and PA of each crop type are better with classification experiment of CNNR_M2-based quasi-Hyperion bands. Very poor PAs are observed in case of Groundnut and Paddy classes because of insufficient samples (i.e., $\leq$30). Wheat class also has very few samples (i.e., $\leq$30), which results to inconsistency in the classification performances of different trials and high SD of UA and PA is observed. We have plotted the average spectral reflectance curves for all the crop types, derived from CNNR_M2-based quasi-Hyperion data, in Fig. 12.

E. Band-Wise Evaluation of Quasi-Hyperion Data and Spectra Comparison With Original Hyperion Data Over the United States Scene

Landsat 8 data are transformed into quasi-Hyperion data for the United States scene. Three models, viz., PHITA, CNNR_M2, and CNNR_M6 are investigated here for

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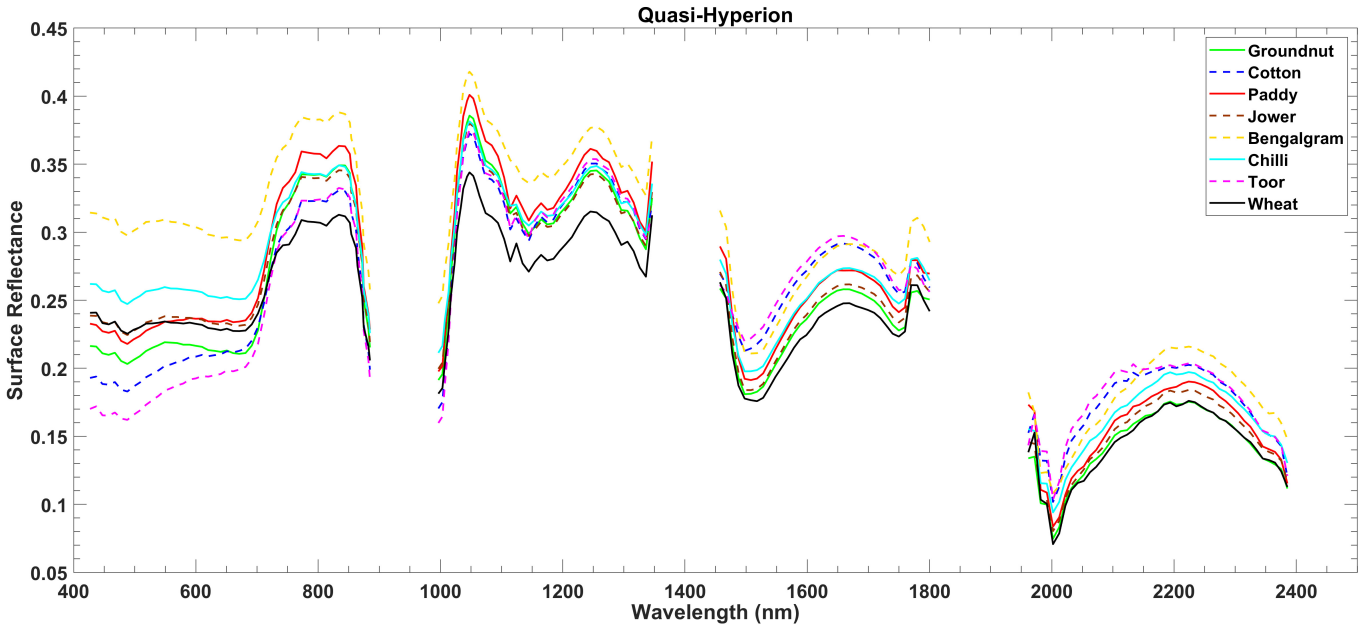


Fig. 12. Average spectral reflectance curves for all the crop types derived from quasi-Hyperion data over the Indian scene.

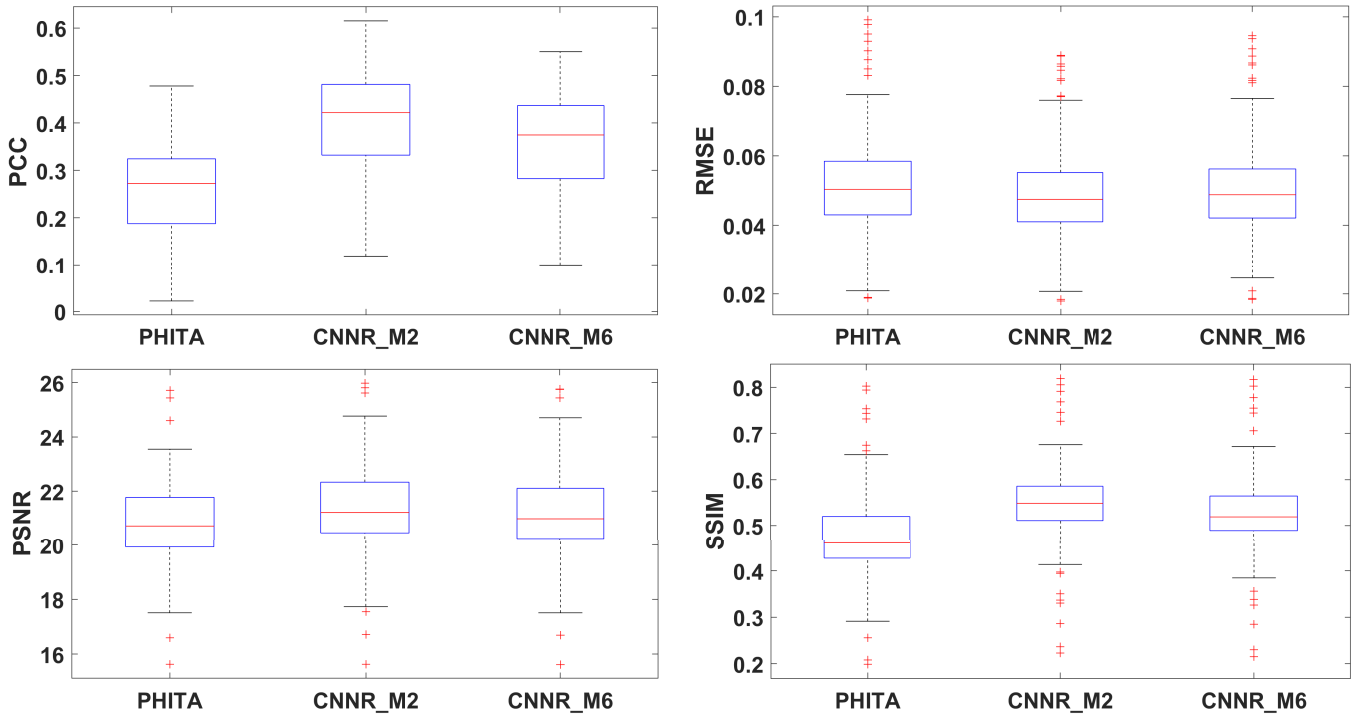


Fig. 13. Comparison of statistical measures (calculated between all the pixels of 170 original and quasi-Hyperion bands) of different models over the United States scene.

this scene. Since PHITA is the existing simple linear model, and among the proposed CNNR architectures, CNNR_M2 is performing better with crop classification application and CNNR_M6 is performing better in terms of statistical metrics for Landsat 7 to quasi-Hyperion transformation over the Indian scene. The statistical metrics (i.e., PCC, RMSE, PSNR, and SSIM) are calculated for each narrow spectral band comparing the reflectance values of original and quasi-Hyperion bands for all the testing pixels. It is observed that CNNR_M2 is performing better consistently with all the statistical metrics (Fig. 13). CNNR_M2 has average improvement of 66.72, 5.57, 2.41, and 15.72% in terms of PCC, RMSE, PSNR, and SSIM, respectively, compared with performance of PHITA. However, the overall performances from different models are not as good as the Landsat 7 transformed quasi-Hyperion data. One possible reason for this poor performance can be unavailability of the both MS and HS images for the same acquisition date.

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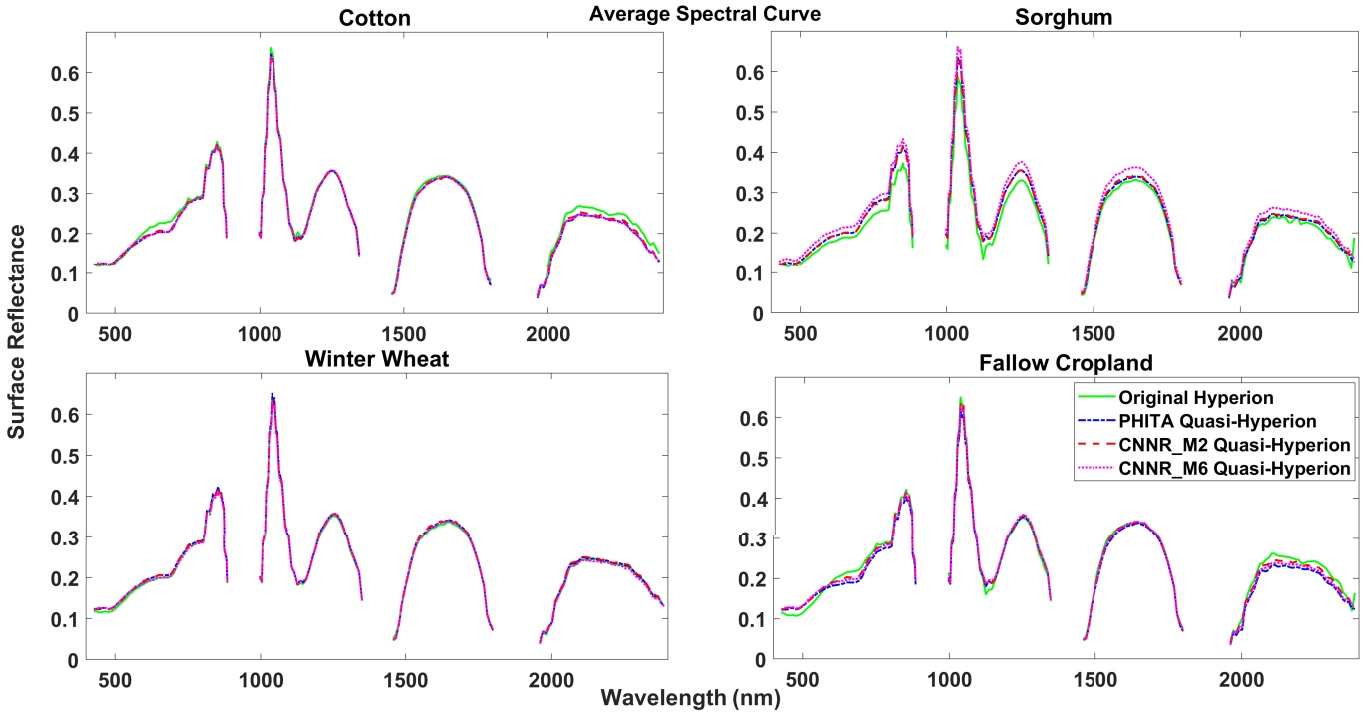


Fig. 14. Comparison between original and different model-based quasi-Hyperion data derived average spectral reflectance curves of four crop classes over the United States scene.

Average spectral reflectance curves of all the four types of crop classes derived from original Hyperion data are compared with the PHITA, CNNR_M2, and CNNR_M6-based quasi-Hyperion data derived crop spectra (Fig. 14). All the models derived from quasi-Hyperion spectra almost look similar to the original Hyperion spectra for all the four crop classes. Hence, SAM is calculated between original and quasi-Hyperion spectra to identify the optimal quasi-Hyperion spectra with minimum error. It is observed that CNNR_M2 derived average spectra have minimum SAM with original spectra for all the four classes, viz., Cotton (SAM: 0.0242), Sorghum (SAM: 0.0373), Winter Wheat (SAM: 0.0236), and Fallow Cropland (SAM: 0.0241).

These computational experiments prove that CNNR_M2 architecture transforms Landsat 8 data into quasi-Hyperion data more efficiently. Indian scene computational experiments also exhibit reasonable performance in terms of statistical measures with CNNR_M2 architecture and optimal performance with crop classification application. Being a simple architecture, CNNR_M2 requires less computational time to train the model. Therefore, CNNR_M2 can be the most suitable one among all the proposed CNNR architectures.

V. Conclusion

In this article, deep learning-based CNNR model is proposed for MS to quasi-HS data transformation. Six different architectures of CNNR model are experimented in order to accomplish best possible construction of quasi-HS data. CNNR models' performances are also compared with the PHITA, SLR, and SVR. All the models are developed and evaluated statistically for an area where both MS (i.e., Landsat 7 ETM+) and HS (i.e., Hyperion) images are available for the same date and almost same acquisition time over an Indian scene. Later, the developed models are employed to create quasi-Hyperion data for an area where original HS images are not available and these quasi-Hyperion data are utilized for crop classification to evaluate the efficiency of the regression models and the transformed data. Landsat 8 to quasi-Hyperion transformation is also carried out over a United States scene with the main objective of crop spectra comparison between original and quasi-Hyperion data. All the CNNR architectures are consistently performing better compared with PHITA, SLR, and SVR with reference to statistical metrics-based evaluation as well as crop classification application. It is observed that CNNR_M6 is performing better with reference to statistical evaluation over the Indian scene but with crop classification application, CNNR_M2 is proven to be more efficient, which is also providing reasonably good performance in terms of statistical metrics for the Indian scene and optimal performance for the United States scene. Therefore, it can be recommended to use the CNNR_M2 model for generation of quasi-HS data, since this architecture is less complex, requires less computational time for training, and has better generalizing capability.

The main limitations of this work are: 1) unavailability of large numbers of reference ground-truth samples with crop type information, which will be very useful for further generalization of crop classification application and 2) different dates of acquisition of the used Landsat 8 and Hyperion data over the United States scene, which may not provide the most efficient performance.

In future, transformation of Landsat 8 or Sentinel 2 bands into quasi-HS bands can be attempted with the availability of HS images for the same area of interest and same date of

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acquisition. Airborne HS data also can play an important role for development of such MS to quasi-HS data generation algorithm. Quasi-HS data can be analyzed for other applications too such as crop condition monitoring, biophysical variables estimation, soil classification, water quality monitoring, and so on. Though quasi-HS data may not replace the need of original HS data but with the development of such efficient data transformation techniques, it can definitely fill the gap till any space-borne HS sensor becomes available globally.

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